

Study Guide: Acids and Bases

This sheet collects the asides you should be able to recall before the assessment. Each box highlights a different kind of point.

DEFINITION

An acid is a substance that donates a proton, while a base is a substance that accepts one. The strength of an acid reflects how completely it dissociates in water.

WARNING

Always add acid to water, never water to acid. The reverse can boil violently and splash concentrated acid out of the container.

TIP

Remember the pH scale runs from 0 to 14, with 7 being neutral. Each whole step is a tenfold change in hydrogen-ion concentration.

More Asides

NOTE

A buffer solution resists changes in pH when small amounts of acid or base are added. Buffers are central to keeping blood chemistry stable.

EXAMPLE

Hydrochloric acid in the stomach has a pH of roughly 1.5 to 3.5, which is acidic enough to begin breaking down the proteins in food.

KEY POINT

Neutralisation between an acid and a base always produces a salt and water. The reaction is exothermic and releases measurable heat.